An hybrid scheme supporting polygonal meshes for elliptic interface problems

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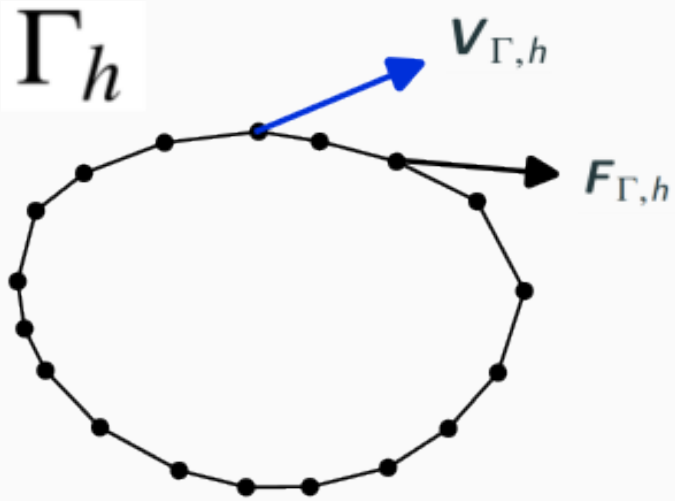
A motivation: meshed internal boundaries

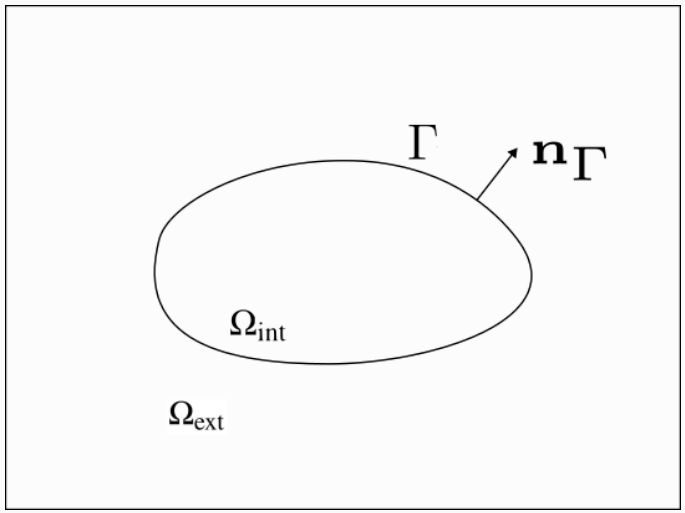
- Reference scenario: flow of a red blood cell, treated with an Immersed Boundary Method [Peskin 2002]
- Goal: include electricity computations, leading to problems with unknowns discontinuous across the interface



Elliptic Interface Problem

Given: conductivity $\sigma : \Omega \setminus \Gamma \to \mathbb{R}$ s.t. $\sigma_{|\Omega_{\text{int}}} \equiv \sigma_{\text{int}} > 0$ and $\sigma_{|\Omega_{\text{ext}}} \equiv \sigma_{\text{ext}} > 0$; $f : \Omega \setminus \Gamma \to \mathbb{R}$, $\Phi_\Gamma : \Gamma \to \mathbb{R}$, and $J_\Gamma : \Gamma \to \mathbb{R}$; find potential $u = (u_{\text{int}}, u_{\text{ext}})$ with $u_\bullet : \Omega_\bullet \to \mathbb{R}$, $\bullet \in \{\text{int}, \text{ext}\}$ such that:

$$
\begin{aligned}
-\nabla \cdot (\sigma_\bullet \nabla u_\bullet) &= f && \text{in } \Omega_\bullet \\
[u]_\Gamma &= J_\Gamma && \text{on } \Gamma \\
[\sigma \nabla u]_\Gamma \cdot n_\Gamma &= \Phi_\Gamma && \text{on } \Gamma \\
u_{\text{ext}} &= 0 && \text{on } \partial\Omega
\end{aligned}
$$



where $[q]_\Gamma := q_{\text{int}} - q_{\text{ext}}$.

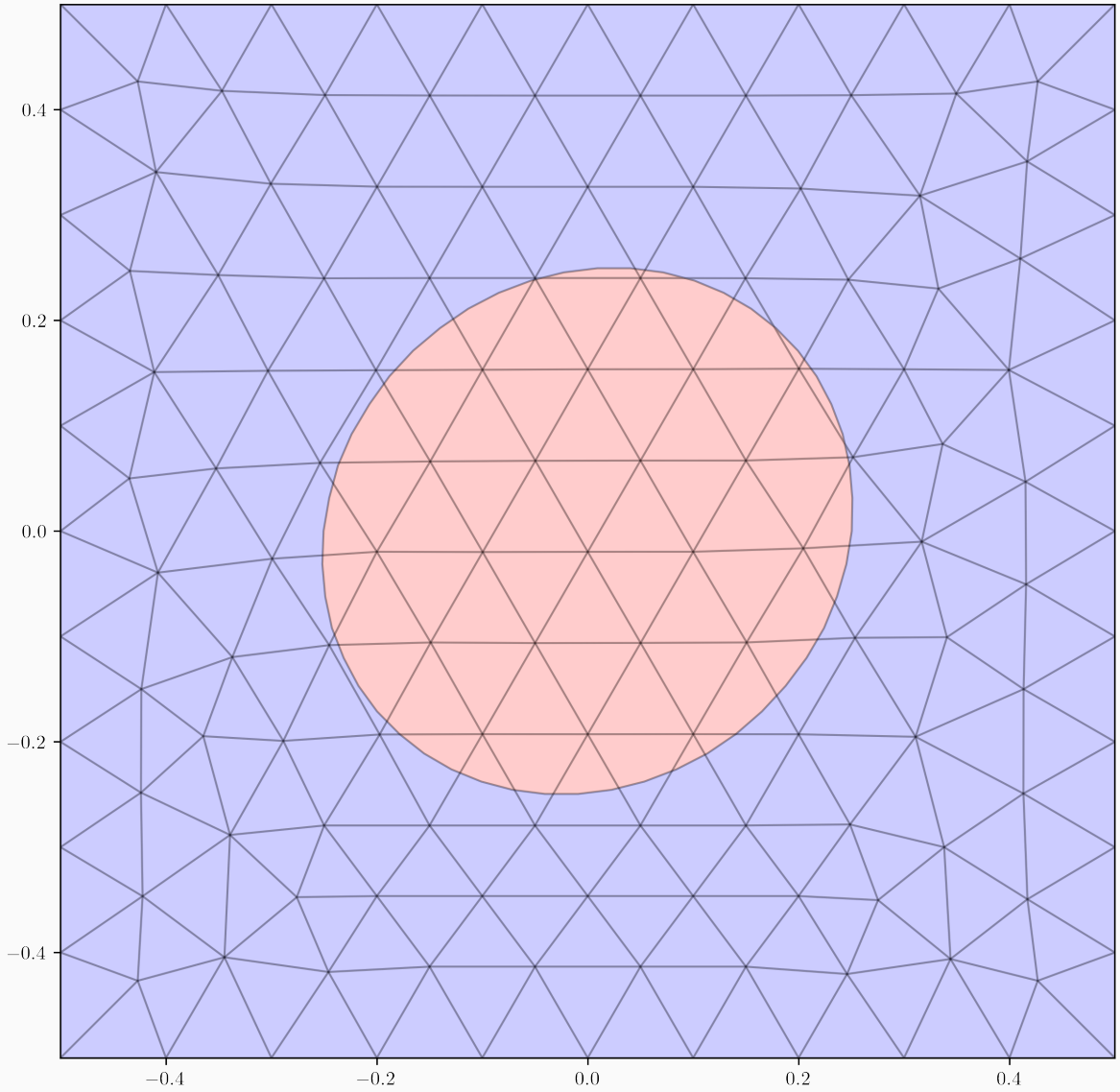
Existing Unfitted Approaches

- FD, FV:
  - Immersed Interface [Li 2003]
  - Ghost Fluid [Fedkiw et al. 1999]
- FEM:
  - XFem [Strouboulis, Babuška, and Copps 2000]
  - Cut-Fem [Burman et al. 2015]
  - ϕ-Fem [Lozinski, Duprez, Lleras]
- Hybrid: Cut-HHO [Burman and Ern 2018]

Typically referred to Level Set discretizations

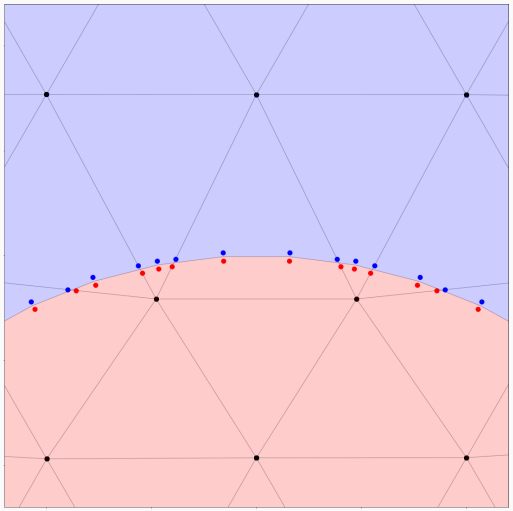
Desired discrete setting

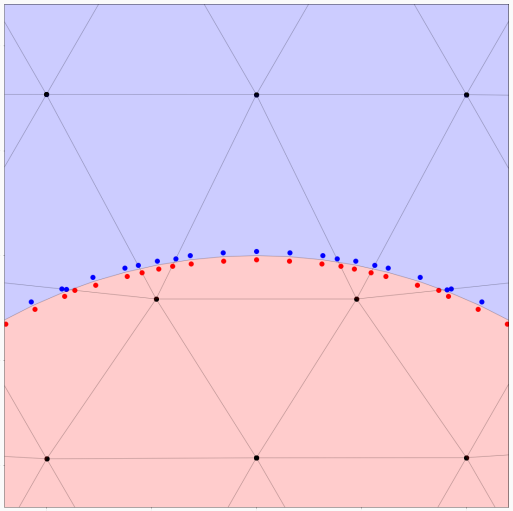
Polygonal Interface Γ_h, mesh $(\mathcal{T}_h, \mathcal{E}_h, \mathcal{V}_h)$, **fitted** to Γ_h



Main Advantages

- High-fidelity representation of the curved interface (refinement of Γ_h independent of background mesh)
- Seamless treatment of cuts (since fitted)





Cornerstones of the Solver

- Unknown the H^1-like hybrid space from [Di Pietro and Droniou 2023, DDR] (polygonal elements)
- Weak imposition of interface condition in the spirit of Nitsche's method [Nitsche 1971]
- Robustness w.r.t. $\frac{\sigma_{\mathrm{int}}}{\sigma_{\mathrm{ext}}}$, leveraging a weighting technique introduced in [Burman and Zunino 2006]

Passage to weak formulation

With $\bullet \in \{\mathrm{int}, \mathrm{ext}\}$ and $v = (v_{\mathrm{int}}, v_{\mathrm{ext}})$ test function:

$$\left(\sum_{\bullet \in \{\mathrm{int},\mathrm{ext}\}} \int_{\Omega_\bullet} \sigma_\bullet \nabla u_\bullet \cdot \nabla v_\bullet \right) - \int_\Gamma \{\sigma_\bullet \nabla u\} \cdot n_\Gamma [v]_\Gamma =$$
$$\left(\sum_{\bullet \in \{\mathrm{int},\mathrm{ext}\}} \int_{\Omega_\bullet} f v_\bullet \right) + \int_\Gamma \Phi_\Gamma \{\sigma_\bullet \nabla u\} \cdot n_\Gamma,$$

using:

$$\int_\Gamma [\sigma \nabla u \cdot n_\Gamma v]_\Gamma = \int_\Gamma \{\sigma \nabla u \cdot n_\Gamma\} [v]_\Gamma + \int_\Gamma [\sigma \nabla u \cdot n_\Gamma]_\Gamma \{v\}.$$

Discrete Space

After [Di Pietro and Droniou 2023] (DDR):
Local DOF space:

$$\underline{V}_T^k = \Big\{ \big(v_T, (v_E)_{E\in\mathcal{E}_T}, (v_V)_{V\in\mathcal{V}_T}\big) \ : \ v_T \in \mathcal{P}^{k-1}(T), \ v_E \in \mathcal{P}^{k-1}(E) \quad \forall E \in \mathcal{E}_T, \ v_V \in \mathbb{R} \quad \forall V \in \mathcal{V}_T \Big\}$$

Global space $\underline{V}_h^k$:

- Glue together DOFs on common edges and vertices
- Keep DOFs doubled on Γ_h

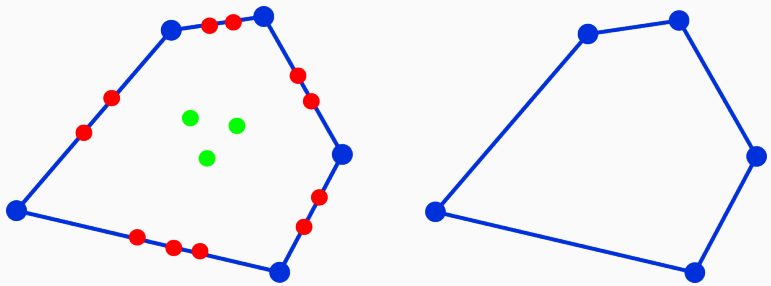


Figure 1: $k = 1$ VS $k = 0$

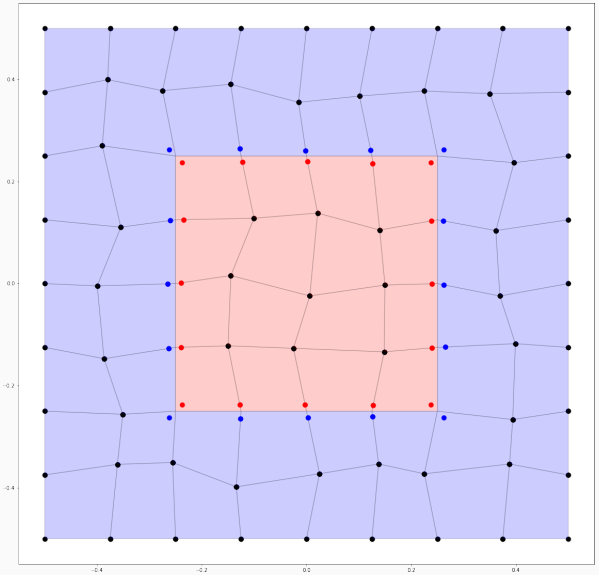


Figure 2: $k = 0$: spots are DOFs

Polinomially Consistent Reconstructions

Fully discrete method: $\underline{V}_h \not\subset L^2(\Omega)$, need to define operators:

- Trace reconstruction $(\cdot)_{TE} : \underline{V}_T^k \to \mathcal{P}^{k+1}(E)$, with $E \in \mathcal{E}_T$
- Gradient reconstruction $G_T^k : \underline{V}_T^k \to \mathcal{P}^k(T)^2$
- Element reconstruction $p_T^{k+1} : \underline{V}_T^k \to \mathcal{P}^{k+1}(T)$

With $w \in \mathcal{P}^{k+1}(T)$, and introducing an interpolator based on L^2 projections and evaluations at nodes:

$$\underline{I}_T^k(w) := \left(\pi_T^{k-1} w, (\pi_E^{k-1} w)_{E \in \mathcal{E}_T}, (w(x_V))_{V \in \mathcal{V}_T}\right),$$

it holds:

- $(\underline{I}_T^k w)_{TE} = w|_E$
- $G_T^k(\underline{I}_T^k w) = \nabla w$
- $p_T^{k+1}(\underline{I}_T^k) w = w$

Definitions of Local Reconstructions

Edge reconstruction $v_{TE} \in \mathcal{P}^{k+1}(E)$ s.t.

$$v_{TE}(x_V) = v_V \text{ for any endpoint } V \text{ of } E \text{ and } \pi_E^{k-1} v_{TE} = v_E;$$

Discrete gradient $G_T^k : \underline{V}_T^k \to \mathcal{P}^k(T)^2$:

$$\int_T G_T^k \underline{v}_T \cdot \tau = -\int_T v_T (\nabla \cdot \tau) + \sum_{E \in \mathcal{E}_T} \omega_{TE} \int_E v_{TE} (\tau \cdot n_E) \quad \forall \tau \in \mathcal{P}^k(T)^2, \tag{1}$$

Element reconstruction $p_T^{k+1} : \underline{V}_T^k \to \mathcal{P}^{k+1}(T)$:

$$\int_T p_T^{k+1} \underline{v}_T (\nabla \cdot \tau) = -\int_T G_T^k \underline{v}_T \cdot \tau + \sum_{E \in \mathcal{E}_T} \omega_{TE} \int_E v_{TE} (\tau \cdot n_E)$$
$$\forall \tau \in \mathcal{R}^{c,k+2}(T). \tag{2}$$

Operators along interface

Edge Jump:

$$[\underline{v}_h]_E := v_{T_{\mathrm{int}}E} - v_{T_{\mathrm{ext}}E}; \tag{3}$$

Average:

$$\lambda_{\mathrm{int}} := \frac{\sigma_{\mathrm{ext}}}{\sigma_{\mathrm{int}} + \sigma_{\mathrm{ext}}}, \qquad \lambda_{\mathrm{ext}} := \frac{\sigma_{\mathrm{int}}}{\sigma_{\mathrm{int}} + \sigma_{\mathrm{ext}}}, \tag{4}$$

with:

$$\{\underline{v}_h\}_{\overline{\lambda},E} := \lambda_{\mathrm{ext}} v_{T_{\mathrm{int}}E} + \lambda_{\mathrm{int}} v_{T_{\mathrm{ext}}E}; \tag{5}$$

Skew Average:

$$\{v\}_{\lambda,E} := \lambda_{\mathrm{int}} \gamma_{T_{\mathrm{int}}E} v + \lambda_{\mathrm{ext}} \gamma_{T_{\mathrm{ext}}E} v. \tag{6}$$

Remark: weights in average operators are pivotal to prove robustness w.r.t. σ constrast.

Discrete Problem

Find $\underline{u}_h \in \underline{V}_{h,0}^k$ s.t.:

$$a_h(\underline{u}_h, \underline{v}_h) = \ell_h(\underline{v}_h) \qquad \forall \underline{v}_h \in \underline{V}_{h,0}^k, \tag{7}$$

with:

$$\begin{aligned} a_h(\underline{w}_h, \underline{v}_h) := & \sum_{T \in \mathcal{T}_h} \left(\int_T \sigma_T G_T^k \underline{w}_T \cdot G_T^k \underline{v}_T + \frac{\sigma_T}{h_T} \sum_{E \in \mathcal{E}_T} \int_E (p_T^{k+1} \underline{w}_T - w_{TE})(p_T^{k+1} \underline{v}_T - v_{TE}) \right) \\ & - \sum_{E \in \mathcal{E}_{\Gamma,h}} \int_E \{\sigma G_h^k \underline{w}_h\}_{\lambda,E} \cdot n_E [\underline{v}_h]_E + \eta \sum_{E \in \mathcal{E}_{\Gamma,h}} \frac{\alpha}{h_E} \int_E [\underline{w}_h]_E [\underline{v}_h]_E, \end{aligned} \tag{8}$$

where

$$\alpha := \frac{2\sigma_{\mathrm{int}} \sigma_{\mathrm{ext}}}{\sigma_{\mathrm{int}} + \sigma_{\mathrm{ext}}}, \tag{9}$$

and

$$\ell_h(\underline{v}_h) := \sum_{T \in \mathcal{T}_h} \int_T f p_T^{k+1} \underline{v}_T + \sum_{E \in \mathcal{E}_{\Gamma,h}} \int_E \Phi_\Gamma \{\underline{v}_h\}_{\overline{\lambda},E} + \eta \sum_{E \in \mathcal{E}_{\Gamma,h}} \frac{\alpha}{h_E} \int_E J_\Gamma [\underline{v}_h]_E. \tag{10}$$

Stability

Proposition [Di Pietro, Mendez, and Spadotto 2025]

Let $N_\partial := \max_{T \in \mathcal{T}_h} \mathrm{card}(\mathcal{E}_T \cap \mathcal{E}_{\Gamma,h})$ and C_{tr} be the trace inequality constant. Assume that

$$\eta > \frac{C_{\mathrm{tr}}^2 N_\partial}{4\epsilon} \tag{11}$$

for some real number $0 < \epsilon < 1$. Then, for all $\underline{v}_h \in \underline{V}_{h,0}^k$, it holds

$$C_{\mathrm{stab}} \|\underline{v}_h\|_{\mathrm{en},h}^2 \leq a_h(\underline{v}_h, \underline{v}_h), \tag{12}$$

with $C_{\mathrm{stab}} := \min\left\{1 - \epsilon, \eta - \frac{C_{\mathrm{tr}}^2 N_\partial}{4\epsilon}\right\}$.

Convergence

Proposition [Di Pietro, Mendez, and Spadotto 2025]

Let $u = (u_{\mathrm{int}}, u_{\mathrm{ext}}) \in \left[C^0(\overline{\Omega}_{\mathrm{int}}) \cap H^{r+2}(\mathcal{T}_h^{\mathrm{int}})\right] \times \left[C^0(\overline{\Omega}_{\mathrm{ext}}) \cap H^{r+2}(\mathcal{T}_h^{\mathrm{ext}})\right]$. Under assumption (11), with $\overline{\sigma} := \max\{\sigma_{\mathrm{int}}, \sigma_{\mathrm{ext}}\}$ and for some $r \in \{0, \dots, k\}$, it holds:

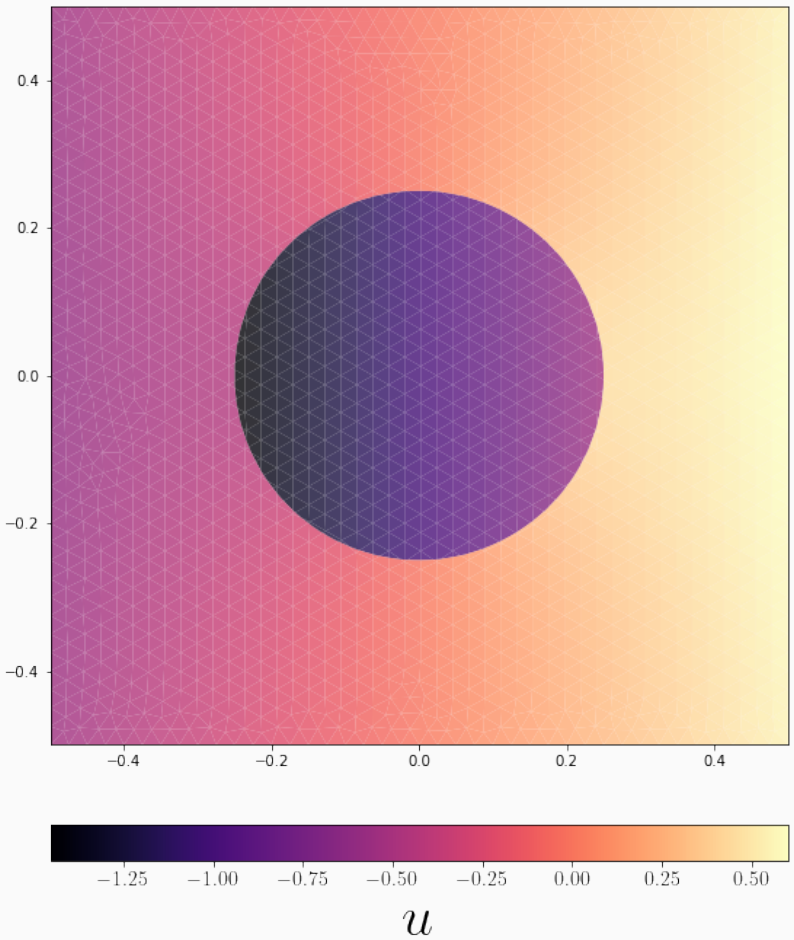
$$\|\underline{u}_h - \underline{I}_h^k u\|_{\mathrm{en},h} \lesssim \overline{\sigma}^{1/2} h^{r+1} |u|_{H^{r+2}(\mathcal{T}_h)},$$

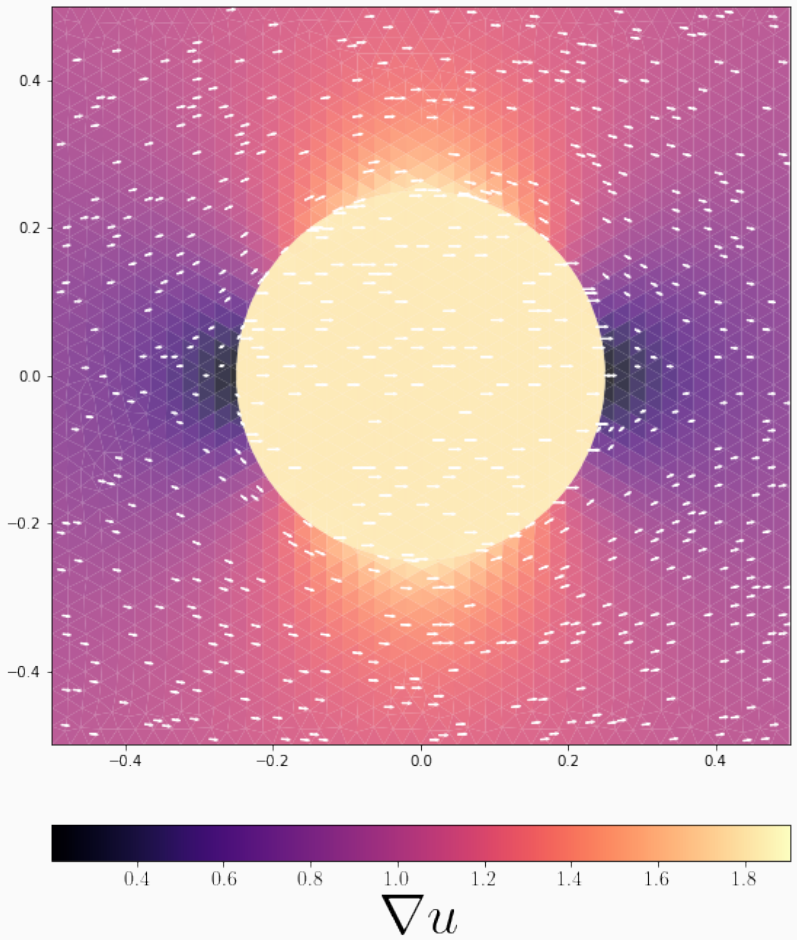
where $|\cdot|_{H^{r+2}(\mathcal{T}_h)}$ is the broken H^{r+2}-seminorm on the mesh $\mathcal{T}_h$ and the hidden constant depends only on the domain, the stability constant C_{stab}, the polynomial degree k, and the mesh regularity parameter (but is independent of both the meshsize and σ).

Numerical Tests

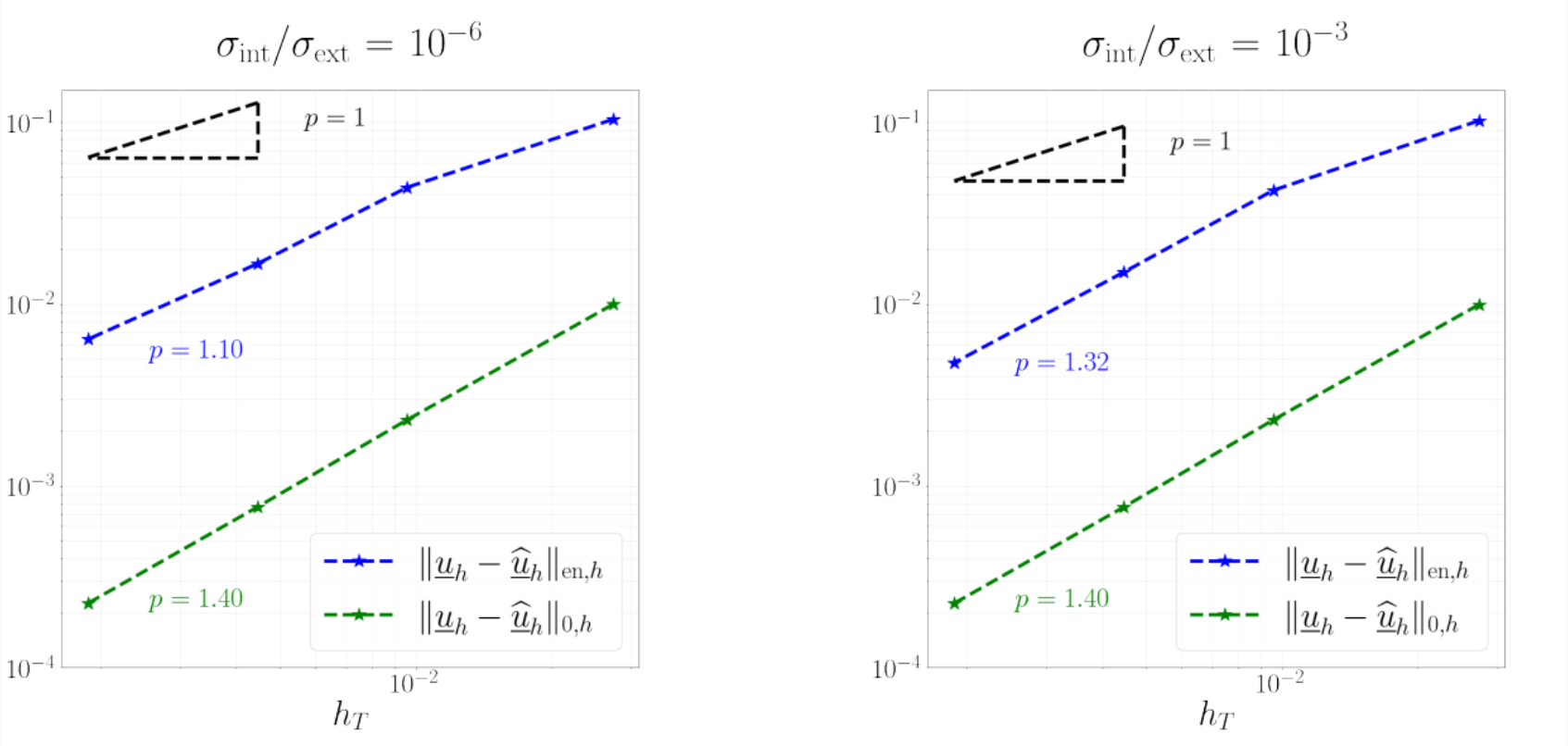
Circular Interface, J_Γ = const

$$u_{\text{int}} = \frac{2\sigma_{\text{ext}}}{\sigma_{\text{ext}} + \sigma_{\text{ext}}} x, \qquad u_{\text{ext}} = 1 + \left[1 + \left(\frac{\sigma_{\text{ext}} - \sigma_{\text{int}}}{\sigma_{\text{ext}} + \sigma_{\text{int}}}\right) \frac{R^2}{x^2 + y^2}\right] x$$



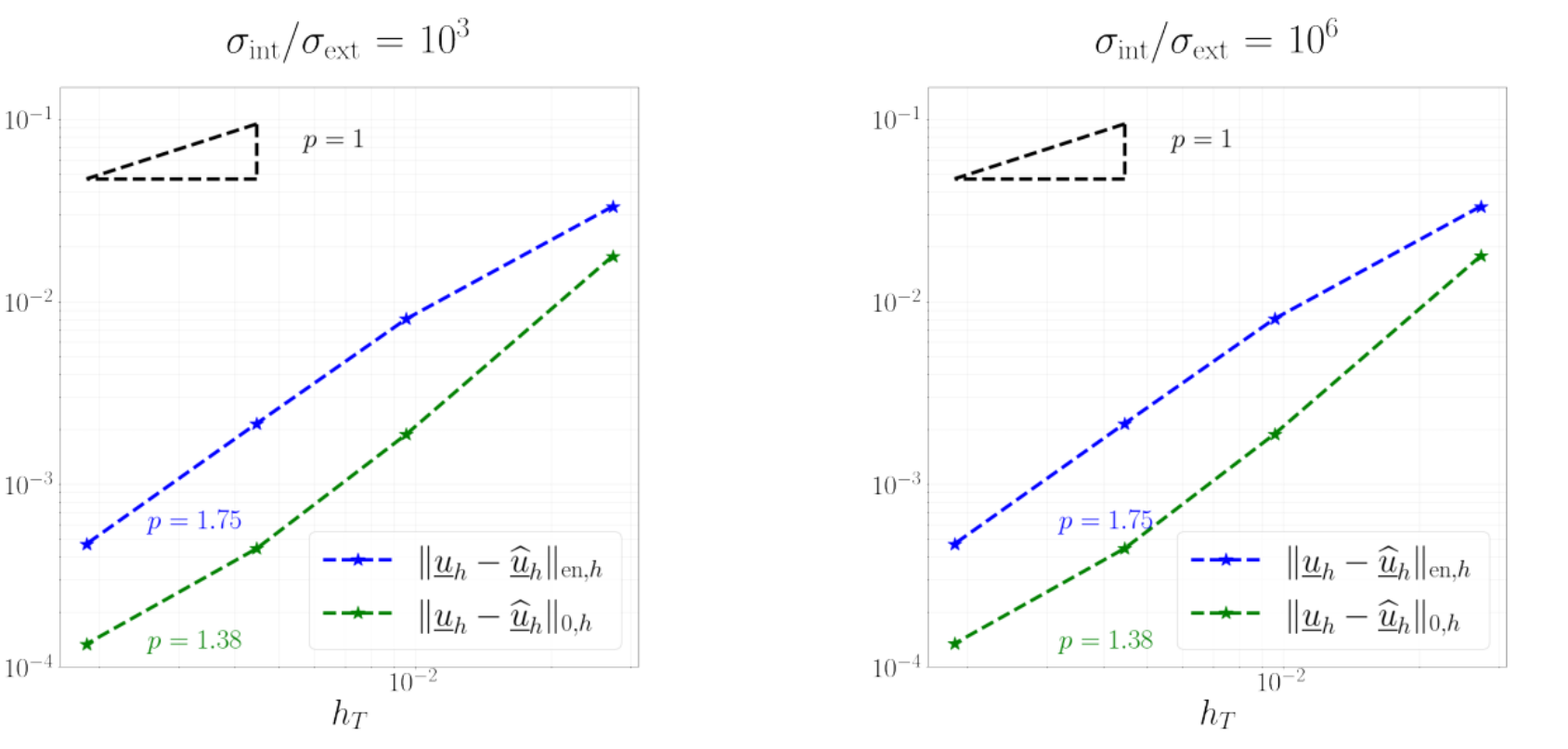


Convergence Trends, $k = 0$



$$\|\underline{v}_h\|_{0,h} := \left(\sum_{T \in \mathcal{T}_h} \|v_T\|^2_{L^2(T)} + h_T \sum_{E \in \mathcal{E}_T} \|v_E\|^2_{L^2(E)} \right)^{\frac{1}{2}} \qquad \forall \underline{v}_h \in \underline{V}_h^k.$$

Convergence Trends, $k = 0$



Time-dependent J_Γ

Given $f : (0, t_\mathrm{f}] \to \mathbb{R}$, J_Γ^0: find $u = (u_\mathrm{int}, u_\mathrm{ext})$ and $J_\Gamma : \Gamma \times [0, t_\mathrm{f}] \to \mathbb{R}$ such that

$$
\begin{aligned}
-\nabla \cdot (\sigma_\bullet \nabla u_\bullet) &= f && \text{in } \Omega_\bullet \times (0, t_\mathrm{f}],\ \bullet \in \{\mathrm{int}, \mathrm{ext}\}, && \text{(13a)}\\
[u]_\Gamma &= J_\Gamma && \text{on } \Gamma \times (0, t_\mathrm{f}], && \text{(13b)}\\
[\sigma \nabla u]_\Gamma \cdot n_\Gamma &= 0 && \text{on } \Gamma \times (0, t_\mathrm{f}], && \text{(13c)}\\
u_\mathrm{ext} &= 0 && \text{on } \partial\Omega \times (0, t_\mathrm{f}] && \text{(13d)}\\
C \partial_t J_\Gamma &= \sigma_\bullet \nabla u_\bullet \cdot n_\Gamma && \text{on } \Gamma \times (0, t_\mathrm{f}],\ \bullet \in \{\mathrm{int}, \mathrm{ext}\} && \text{(13e)}\\
J_\Gamma(\cdot, 0) &= J_\Gamma^0 && \text{on } \Gamma, && \text{(13f)}
\end{aligned}
$$

with $C > 0$.

Discretization of Jump Evolution

Time discretization of (13e):

$$\frac{C}{\tau}\left(J_\Gamma^{n+1} - J_\Gamma^n\right) = \sigma_{\mathrm{int}} \nabla u_{\mathrm{int}}^n \cdot n_\Gamma = \sigma_{\mathrm{ext}} \nabla u_{\mathrm{ext}}^n \cdot n_\Gamma .$$

Space of Jumps:

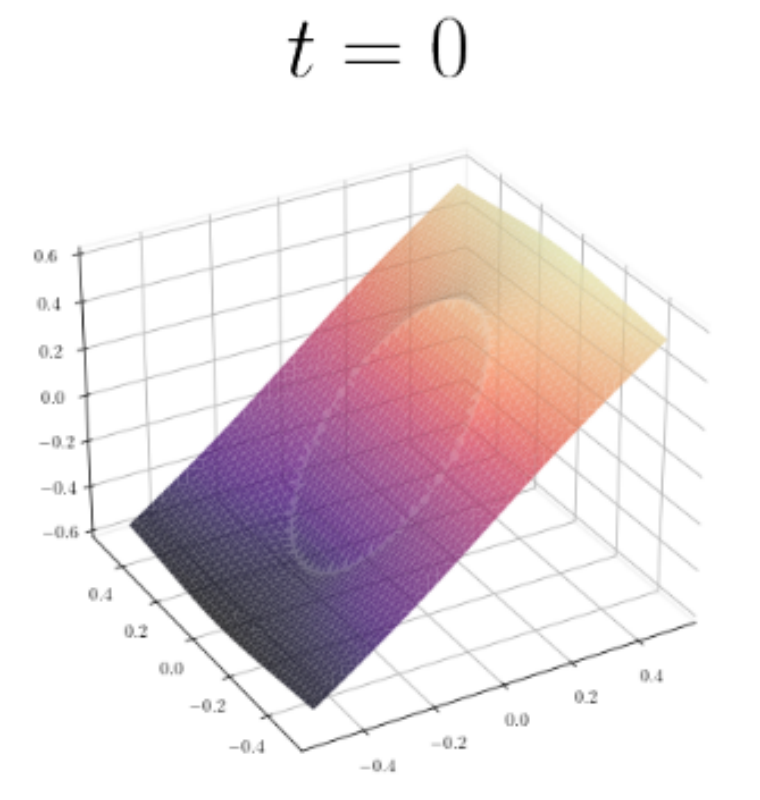
$$\underline{Z}_{\Gamma,h} := \left\{ \underline{p}_{\Gamma,h} = (p_V)_{V \in \mathcal{V}_{\Gamma,h}} \ : \ p_V \in \mathbb{R} \quad \forall V \in \mathcal{V}_{\Gamma,h} \right\}.$$

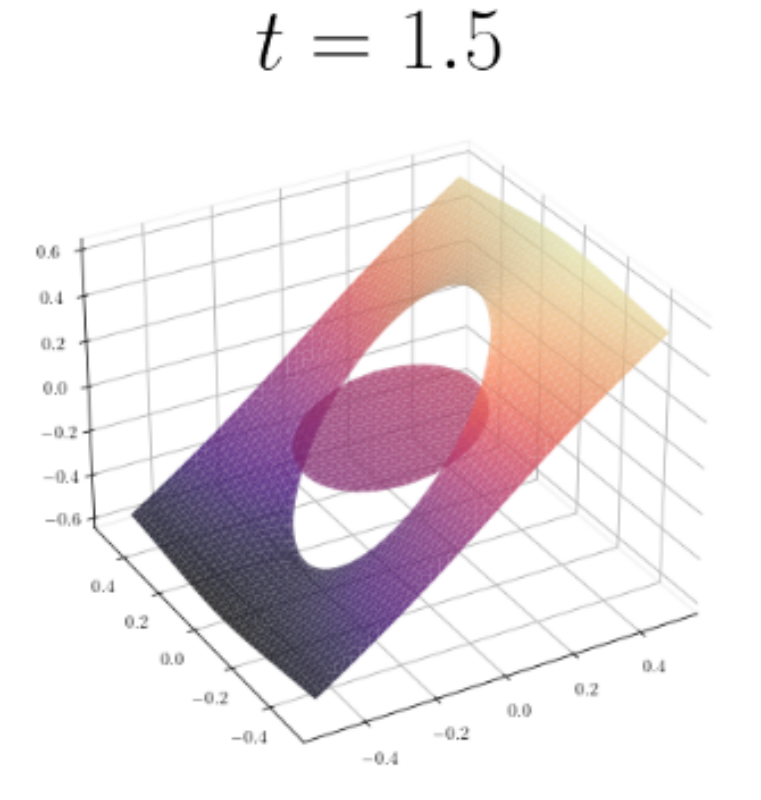
Advancement of Jump: find $\underline{J}_{\Gamma,h}^{n+1} \in \underline{Z}_{\Gamma,h}$ such that

$$\frac{C}{\tau}\left(\sum_{E \in \mathcal{E}_{\Gamma,h}} \int_E J_E^{n+1} Q_E^n - \int_E J_E^n Q_E^n \right) = \sum_{E \in \mathcal{E}_{\Gamma,h}} \int_E \{\sigma G_T \underline{u}_T^n\}_{\lambda,E} \cdot n_\Gamma Q_E \qquad \forall \underline{Q}_{\Gamma,h}^n \in \underline{Z}_{\Gamma,h}. \tag{14}$$

Numeric Test

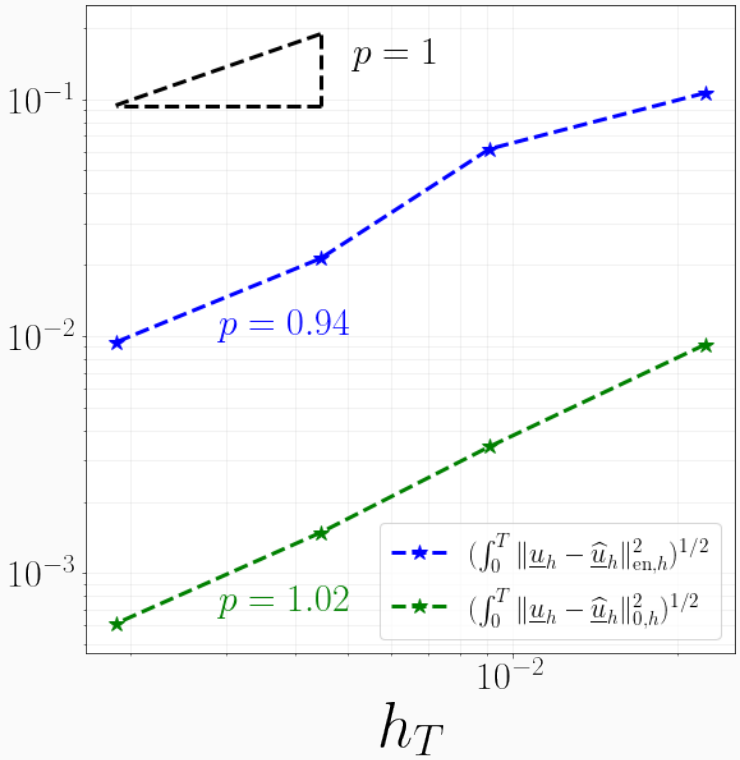
$$\widehat{u}_\bullet(x,t) = \exp\left(-\frac{t}{t_c}\right)\left(u_\bullet^0(x) - u_\bullet^\infty(x)\right) + u_\bullet^\infty(x) \tag{15}$$

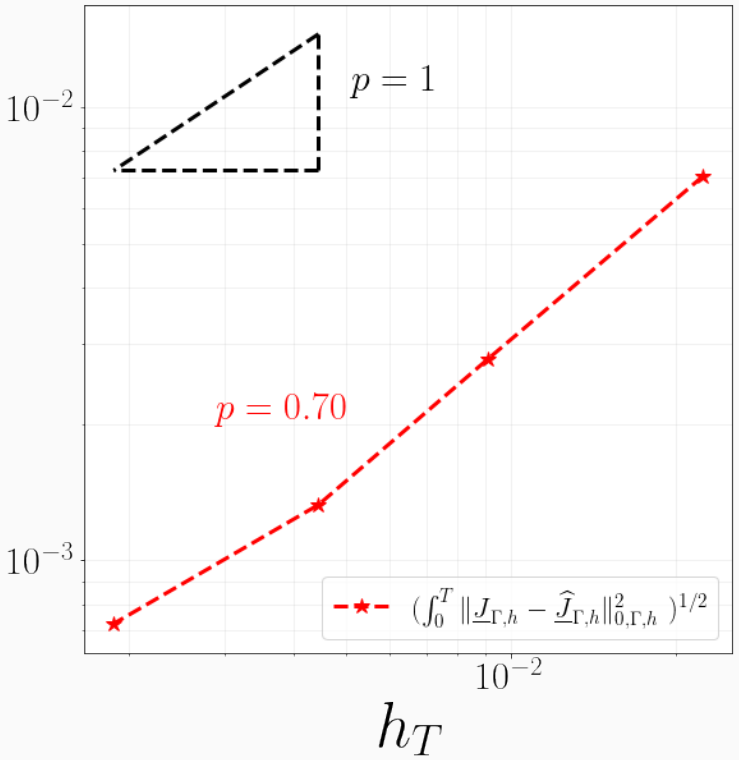




Convergence Trend, $k = 0$

$$\|J_{\Gamma,h}\|^2_{0,\Gamma,h} := \sum_{E \in \mathcal{E}_{\Gamma,h}} \|J_E\|^2_E.$$





Conclusion

- A fitted DDR scheme for an elliptic interface problem
- Arbitrary precision in interface refinement w.r.t. the background mesh
- Stability and convergence regardless of σ contrast

Thank you!

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Backup Material

Stability: Sketch of Proof (I)

$$
\begin{aligned}
a_h(\underline{v}_h, \underline{v}_h) \coloneqq & \sum_{T \in \mathcal{T}_h} \left(\int_T \sigma_T G_T^k \underline{v}_T \cdot G_T^k \underline{v}_T + \frac{\sigma_T}{h_T} \sum_{E \in \mathcal{E}_T} \int_E (p_T^{k+1} \underline{v}_T - v_{TE})(p_T^{k+1} \underline{v}_T - v_{TE}) \right) \\
& - \sum_{E \in \mathcal{E}_{\Gamma,h}} \int_E \{\sigma G_h^k \underline{v}_h\}_{\lambda,E} \cdot n_E [\underline{v}_h]_E + \eta \sum_{E \in \mathcal{E}_{\Gamma,h}} \frac{\alpha}{h_E} \int_E [\underline{v}_h]_E [\underline{v}_h]_E \\
= & \sum_{T \in \mathcal{T}_h} \sigma_T (\| G_T^k \underline{v}_T \|^2_{L^2(T)^2} + h_T^{-1} \sum_{E \in \mathcal{E}_T} \| p_T^{k+1} \underline{v}_T - v_{TE} \|^2_{L^2(E)}) \\
& - \sum_{E \in \mathcal{E}_{\Gamma,h}} \int_E \{\sigma G_h^k \underline{v}_h\}_{\lambda,E} \cdot n_E [\underline{v}_h]_E + \eta |\underline{v}_h|^2_{\mathrm{J},h},
\end{aligned}
\tag{16}
$$

Recall definition of norm:

$$
\| \underline{v}_h \|^2_{\mathrm{en},h} \coloneqq \sum_{T \in \mathcal{T}_h} \sigma_T \left(\| G_T^k \underline{v}_T \|^2_{L^2(T)^2} + h_T^{-1} \sum_{E \in \mathcal{E}_T} \| p_T^{k+1} \underline{v}_T - v_{TE} \|^2_{L^2(E)} \right) + |\underline{v}_h|^2_{\mathrm{J},h}, \tag{17}
$$

Stability: Sketch of Proof (II)

$$
\begin{aligned}
&\int_E \{\sigma G_h^k \underline{w}_h\}_{\lambda,E} \cdot n_E [\underline{v}_h]_E \\
&\leq \|\{\sigma G_h^k \underline{w}_h\}_{\lambda,E}\|_{L^2(E)^2} \|n_E\|_{L^\infty(E)^2} \|[\underline{v}_h]_E\|_{L^2(E)} \\
&\leq \alpha^{-1/2} \left(\lambda_{\mathrm{int}} \sigma_{\mathrm{int}}^{1/2} \|\sigma_{\mathrm{int}}^{1/2} G_{T_{\mathrm{int}}}^k \underline{w}_{T_{\mathrm{int}}}\|_{L^2(E)^2} + \lambda_{\mathrm{ext}} \sigma_{\mathrm{ext}}^{1/2} \|\sigma_{\mathrm{ext}}^{1/2} G_{T_{\mathrm{ext}}}^k \underline{w}_{T_{\mathrm{ext}}}\|_{L^2(E)^2}\right) \ \alpha^{1/2} \|[\underline{v}_h]_E\|_{L^2} \\
&\leq \frac{1}{\sqrt{2}} \left(\|\sigma_{\mathrm{int}}^{1/2} G_{T_{\mathrm{int}}}^k \underline{w}_{T_{\mathrm{int}}}\|_{L^2(E)^2} + \|\sigma_{\mathrm{ext}}^{1/2} G_{T_{\mathrm{ext}}}^k \underline{w}_{T_{\mathrm{ext}}}\|_{L^2(E)^2}\right) \ \alpha^{1/2} \|[\underline{v}_h]_E\|_{L^2(E)} \\
&\leq \frac{C_{\mathrm{tr}}}{\sqrt{2}} \left(\|\sigma_{\mathrm{int}}^{1/2} G_{T_{\mathrm{int}}}^k \underline{w}_{T_{\mathrm{int}}}\|_{L^2(T_{\mathrm{int}})^2} + \|\sigma_{\mathrm{ext}}^{1/2} G_{T_{\mathrm{ext}}}^k \underline{w}_{T_{\mathrm{ext}}}\|_{L^2(T_{\mathrm{ext}})^2}\right) \ \left(\frac{\alpha}{h_E}\right)^{1/2} \|[\underline{v}_h]_E\|_{L^2(E)},
\end{aligned}
$$

exploiting:

$$
\frac{\lambda_\bullet^2 \sigma_\bullet}{\alpha} < \frac{1}{2} \qquad \bullet \in \{\mathrm{int}, \mathrm{ext}\}. \tag{18}
$$